

Polyplex Lie Geometry

A Multi-Metric Complex Geometry Structured by Lie Algebras in a Hilbertian Framework

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Abstract

Polyplex Lie Geometry is a high-dimensional complex geometric framework designed to encapsulate multiple geometric forms, spherical, ellipsoidal, hyperbolic, lattice (cubic), and taxicab, within a unified algebraic and analytic structure. This geometry is embedded in a complex Hilbert space $\mathcal{H}_{\mathbb{C}}^n$, where $n \in \{3, \dots, 6\}$, and admits a rich tensor and operator algebra, enabling it to represent both classical geometric structures and quantum field theoretic symmetries.

At its core, the geometry is governed by Lie algebras, geometric algebra (Clifford algebra), and tensorial outer products, making it compatible with the formalism of gauge theory, general relativity, and non-commutative quantum geometry.

Polyplex Lie Geometry is a high-dimensional complex geometric framework designed to encapsulate and interrelate multiple geometric forms, spherical, ellipsoidal, hyperbolic, lattice (cubic), and taxicab, within a unified algebraic and analytic structure. This geometry is embedded in a complex Hilbert space $\mathcal{H}_{\mathbb{C}}^n$, where $n \in \{3, \dots, 6\}$, where $n \in \{3, \dots, 6\}$, enabling a fusion of continuous and discrete geometries. The space supports a rich tensor and operator algebra, making it compatible with both classical geometric constructions and the symmetries of quantum field theories.

At its foundation, Polyplex Lie Geometry is governed by the interplay between Lie algebras, geometric algebra (Clifford algebra), and tensorial outer product structures, permitting a highly expressive mathematical language suited to gauge theory, general relativity, and non-commutative quantum geometry. The framework integrates inner and outer products, multivector extensions, and matrix representations of bivector generators to encode symmetries and transformations.

The framework begins with a central mathematical singularity, interpreted as a maximally degenerate algebraic point. From this origin, the geometry expands outward to capture increasing degrees of metric structure, and inward toward algebraic minimality, producing a dual-gradient of curvature and algebraic complexity. This expansion is both spatial and metric, as differentiable manifolds with distinct curvature (e.g., Euclidean, spherical, hyperbolic) emerge contextually as local domains within the Hilbert space. These domains admit layered or intersecting metrics, as described by their embedding in the full geometric algebra $\mathcal{C}\ell_n$.

Complementing this continuous structure is the MegaCube, a fundamentally discrete, combinatorially structured lattice constructed from explicit 3×3 complex matrix generators representing geometric bivectors. The MegaCube operates over integer-indexed coordinates, forming a hypercubic grid whose nodes correspond to algebraic configurations in $\mathcal{H}_{\mathbb{C}}^3$. These matrix generators encode directional symmetries and form a closed Lie algebra under commutation, with internal subalgebras isomorphic to $\mathfrak{su}(2)$ and higher structures such as $\mathfrak{so}(6) \approx \mathfrak{su}(4)$.

The critical innovation lies in the intersection of the continuous Lie geometry with the discrete MegaCube lattice, yielding a quantum-consistent interface where smooth vector fields discretize into combinatorial structures. These intersections give rise to quantized geometric states, reflecting the granular character of physical quanta, and are realized at algebraically coherent junctions. The result is a phenomenological geometry that unifies continuous symmetry spaces with lattice-structured quantum granularity, much like the dual role played by spacetime manifolds and Fock spaces in quantum field theory.

The Polyplex framework thus embodies a geometric duality, where the continuous differentiable Lie geometry acts as a dynamic field manifold, while the discrete MegaCube represents a quantized structural skeleton. Together, they offer a topology and algebra capable of encoding rotations, translations, modular symmetries (e.g., $\mathbb{Z}_3$ cyclic generators), and multi-metric embeddings across dimensions. Within this architecture, operator algebras, gauge symmetries, and fiber bundle structures (including Clifford and non-Abelian gauge bundles) emerge naturally, making Polyplex Lie Geometry a fertile mathematical foundation for describing high-dimensional, field-theoretic, and quantum-mechanical systems.

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I. Polyplex Lie Geometry - Key Mathematical Structure

We introduce the foundational framework of Polyplex Lie Geometry, a complex Hilbert space–based formalism that unifies geometric algebra, operator theory, and Lie group symmetries into a single, coherent mathematical structure. Developed to accommodate both classical and quantum systems, the framework is inherently compatible with non-commutative observables and supports a layered approach to geometry, integrating Euclidean, spherical, hyperbolic, ellipsoidal, and discrete lattice metrics. Central to its design is a tensorial and operator-valued architecture that mirrors the structure of gauge fields and quantum mechanics, enabling the encoding of physical laws, symmetry operations, and topological interactions. By allowing intersecting metric domains and dualities between continuous and discrete spaces, Polyplex Lie Geometry offers a robust and flexible model for describing complex physical phenomena across multiple geometric and algebraic dimensions.

i. Complex Hilbert Space Foundation

Let

$$\mathcal{H}_{\mathbb{C}}^n = \text{Hilbert space over } \mathbb{C}, \quad \dim \mathcal{H} = n, \quad 3 \leq n \leq 6$$

with a sesquilinear inner product

$$\langle \psi, \phi \rangle: \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}, \quad \langle \psi, \phi \rangle = \sum_i \psi_i \bar{\phi}_i.$$

ii. Geometric Product Structure

Vectors $a, b \in \mathcal{H}_{\mathbb{C}} \subset \mathbb{C}\ell_n$ obey:

$$ab = a \cdot b + a \wedge b$$

where:

- $a \cdot b = 1/2 (ab + ba)$ is the inner product (scalar-valued)
- $a \wedge b = 1/2 (ab - ba)$ is the bivector (plane segment or Lie algebra generator)

This defines the geometric product consistent with

$$ab = \frac{1}{2}\{a, b\} + \frac{1}{2}[a, b].$$

iii. Lie Algebra Structure

Let $\mathfrak{g} \subset \mathfrak{gl}(n, \mathbb{C})$ be a Lie algebra of linear operators acting on $\mathcal{H}$, with basis elements $\{T_a\}$ satisfying

$$[T_a, T_b] = if_{ab}^c T_c,$$

where $f_{ab}^c \in \mathbb{C}$ are the structure constants. Examples include:

- Electroweak sector: $\mathfrak{g} = \mathfrak{u}(1) \oplus \mathfrak{su}(2)$
- Color sector: $\mathfrak{su}(3)$
- Extended symmetry: $\mathfrak{su}(4)$, $\mathfrak{so}(6, \mathbb{C})$ or higher-dimensional real forms

These Lie algebras generate non-commutative operator algebras on $\mathcal{H}_{\mathbb{C}}^n$ which capture the symmetries and conservation laws relevant to physical field theories.

iv. Polyplex Geometry: Metric Layering

Polyplex Lie Geometry accommodates several metric geometries, superimposed or contextually applied, using piecewise or domain-specific metrics:

a) *Euclidean (flat curvature):*

$$ds^2 = \delta_{ij} dx^i dx^j,$$

b) *Ellipsoidal (Lamé coefficients $H_{\lambda, \mu, \nu}$):*

$$ds^2 = H_{\lambda}^2 d\lambda^2 + H_{\mu}^2 d\mu^2 + H_{\nu}^2 d\nu^2$$

c) *Spherical (positive curvature):*

$$ds^2 = R^2 (d\theta^2 + \sin^2(\theta) d\phi^2 + \dots)$$

d) *Hyperbolic (negative curvature):*

$$ds^2 = R^2 (d\chi^2 + \sinh^2(\chi) d\Omega^2 + \dots)$$

e) *Axial Light (null basis/cone):*

$$ds^2 = 2\alpha(dx^0 dx^1 + dx^2 dx^3 + \dots) = 0, \text{ for } \alpha = e^{(\chi \pm i\pi n/4)}, \chi \in \mathbb{R}$$

f) *Cubic lattice (discrete topological):* Integer-coordinated grid with edgewise adjacency; compatible with graph Laplacians or $\mathbb{Z}^2$ embeddings.

g) *Taxicab (ℓ_1 norm):*

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

Each of these geometries can be interpreted within subdomains of $\mathcal{H}_{\mathbb{C}}^n$ or as quotient spaces of the full geometric algebra, allowing layered or intersecting domain-specific metrics.

v. Tensor and Outer Product Structure

The space includes rank-2 and higher tensors through outer products:

$$A \otimes B \in \text{End}(\mathcal{H}) \otimes \text{End}(\mathcal{H}), \quad A, B \in \mathfrak{g}$$

which define operator-valued fields, multivectors, or connection forms (e.g., in gauge theories). These extend naturally into:

- Fiber bundles over the geometric base space
- Clifford bundles when incorporating spinor structure
- Non-abelian gauge field representations

vi. Operator Algebra and Quantum Compatibility

All observables $\hat{O}_i \in \text{End}(\mathcal{H}_{\mathbb{C}}^n)$ belong to operator algebras where

$$[\hat{O}_i, \hat{O}_j] = i\hbar [\widehat{O_i, O_j}].$$

This framework naturally accommodates:

- Heisenberg algebra (canonical quantization)
- Gauge algebra representations
- Quantum field observables

The structure is fully quantum compatible, with commutators defining time evolution and symmetry generators, and the inner product preserving probability amplitudes. Below, in **Figure 1** we show a plotted image that depicts the layering of multiple geometric forms in one Hilbert space over $\mathbb{C}(\mathcal{H}_{\mathbb{C}}^6)$, where the X_i, iX_j -axes are complex valued.

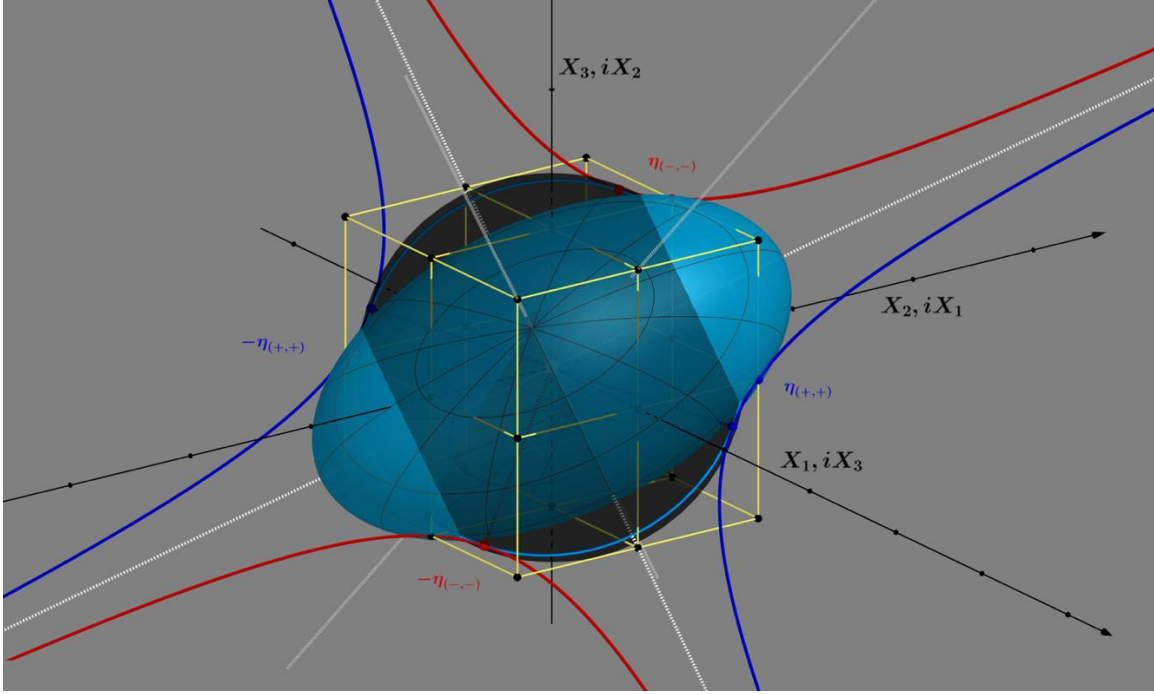


Figure 1. – Polyplex Lie Geometry image $\in \mathcal{H}_{\mathbb{C}}^3$, showing a spherical singularity (black), the cubic lattice structure (yellow), a slice from a quad-hyperboloid geometry (red/blue hyperbolae), and an ellipsoid transform of the singularity, which intersects mid-points on four of the cube's edges (light blue).

Polyplex Lie Geometry is a complex Hilbertian operator geometry of dimension $n = 6 - 12$, equipped with a geometric algebra, embedded Lie algebra $\mathfrak{g} \subset \mathfrak{gl}(n, \mathbb{C})$, tensorial outer product structure, and a layered metric formalism combining continuous and discrete geometries. It is suitable for modeling physical systems with both commuting and non-commuting observables, geometric duality, and non-trivial topological structure.

II. Algebraic Unification with Lie Geometry: Quaternions, Geometric Algebra, and $SU(2)$

1. Introduction and Background

Quaternions, originally introduced by Hamilton, form a non-commutative extension of complex numbers and provide a powerful mathematical tool for representing three-dimensional rotations and orientations. However, despite their versatility, quaternions have traditionally lacked a comprehensive algebraic basis similar to geometric algebra (GA). The present framework integrates quaternion algebra into the richer structure of geometric algebra and clarifies their deep relationship with the Lie group $SU(2)$, particularly through the Pauli matrices.

2. Quaternion Matrices and Pauli Algebra

The quaternions $\mathbb{H}$ can be represented using the following matrix forms

$$i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad j = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad k = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

These quaternionic matrices faithfully replicate Hamilton's quaternionic relations

$$i^2 = j^2 = k^2 = ijk = -I.$$

Each quaternionic unit squares explicitly to the negative identity matrix $-I$, consistent with quaternionic multiplication. Moreover, quaternionic multiplication is anti-commutative

$$ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j.$$

3. Geometric Algebra Basis and Pauli Matrices

The Pauli matrices explicitly form a basis for the Lie algebra $\mathfrak{su}(2)$, defined as

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices satisfy the Lie algebra commutation relations

$$[\sigma_x, \sigma_y] = 2i\sigma_z, [\sigma_y, \sigma_z] = 2i\sigma_x, [\sigma_z, \sigma_x] = 2i\sigma_y.$$

Addressing the limitation of quaternions lacking a defined algebraic basis akin to geometric algebra, the Polyplex Geometric Lie Algebra (PGL) introduces the following quaternionic basis explicitly

$$\{1, i, j, k\}.$$

Normally, the identity matrix $\notin \mathfrak{su}(2)$, since it is not traceless. Therefore, the span of the products of Pauli matrices is larger than $\mathfrak{su}(2)$, generating the $\mathbb{C}\ell_3(\mathbb{C})$ Clifford algebra. Relating to the algebraic structure, we identify quaternionic units through products of the standard Pauli matrices $\sigma_x, \sigma_y, \sigma_z$:

$$i \Leftrightarrow \sigma_x \sigma_y, \quad j \Leftrightarrow \sigma_y \sigma_z, \quad k \Leftrightarrow \sigma_z \sigma_x.$$

The quaternionic relations can also be expressed in the more standard physics convention as

$$i = i\sigma_z, \quad j = i\sigma_y, \quad k = i\sigma_x.$$

4. Unification of Quaternions with Geometric Algebra

The algebraic unification is accomplished by embedding quaternionic algebra into geometric algebra (GA) through defined bivector basis elements, leveraging the established relations

$$i \equiv e_1 e_2, \quad j \equiv e_3 e_2, \quad k \equiv e_3 e_1,$$

where $e_j e_k$ represent oriented bivectors in geometric algebra. This mapping explicitly connects quaternionic units to geometric algebra bivectors, revealing an intrinsic Lie algebraic structure.

Lie Algebra and Quaternionic Algebra

The quaternionic bivectors explicitly form the Lie algebra $\mathfrak{su}(2)$. Using the Lie algebra structure, we define the isomorphisms:

$$\mathfrak{su}(2) \cong \text{span}_{\mathbb{R}}\{i\sigma_x, i\sigma_y, i\sigma_z\},$$

with the commutation relations

$$[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c.$$

where ϵ_{abc} is the Levi-Civita antisymmetric tensor. Thus, the quaternionic algebra is explicitly isomorphic to the Lie algebra $\mathfrak{su}(2)$, enabling direct translation of quaternionic statements into Lie algebraic language. The quaternion Lie algebraic element can therefore be represented as

$$X = x^1 i\sigma_x + x^2 i\sigma_y + x^3 i\sigma_z.$$

for $x^i \in \mathbb{R}$, directly analogous to the Lie algebra representation of $\mathfrak{su}(2)$.

5. Applications

This unified algebraic framework provides a powerful toolkit for theoretical physics, especially in quantum mechanics, quantum information theory, and relativistic formulations where $SU(2)$ symmetry and spinor representations naturally arise. The direct linkage between quaternions, geometric algebra, and Lie algebraic structures underscores their fundamental importance and facilitates clearer mathematical modeling and interpretation of physical phenomena. By explicitly constructing quaternionic algebra through the geometric algebraic basis and connecting them rigorously to the Lie algebra $\mathfrak{su}(2)$ represented via Pauli matrices, this framework achieves a robust algebraic and geometric unification. It allows seamless transitions between quaternionic, geometric algebraic, and Lie algebraic perspectives, significantly enhancing analytical versatility and physical insight.

III. The Polyplex Geometry of the MegaCube

The Polyplex Geometry of the MegaCube is constructed on a foundation of explicit matrix generators, each defined as complex valued 3×3 matrices. We introduce X_1, X_2 , and X_3 -oriented generators, denoted as $E^{X_1}, E^{X_2}, E^{X_3}$, which are representative of bivector elements in geometric algebra and act as foundational elements in the MegaCube algebra. These generators encode the fundamental directional symmetries and commutation structures of the space and form a closed algebra under Lie bracket operations.

The MegaCube itself is defined as a fundamentally discrete space: a quantized lattice of algebraic points forming a hypercubic structure that extends across an abstract manifold. Each node of this lattice corresponds to a matrix-state configuration, and the entire structure evolves combinatorially through algebraic actions of the generator set. In contrast, the Polyplex Lie geometry emerges as a continuous manifold, beginning from a central mathematical singularity, where the algebraic content is maximally degenerate, and expanding both outward and inward from this point. This expansion is not merely spatial but also metric: various metric spaces (Euclidean, hyperbolic, and others) emerge as differentiable regions within the Polyplex structure, guided by transformations and curvature induced by the Lie algebraic flow.

Importantly, the continuous Lie geometry intersects with the discrete MegaCube lattice at specific algebraically coherent junctions. These intersections encode a quantized interface between continuum and discretum, consistent with the phenomenology of quanta in physical theory. At these points, the smooth vector fields of the Lie geometry discretize into finite combinatorial actions, corresponding to observable quantum events. The Polyplex framework thus unifies the smooth symmetries of Lie groups with the granular structure of the MegaCube, creating a novel mathematical topology in which discrete quantum phenomena are embedded in, and emergent from, a deeper geometric continuity.

1. Matrix Generator Definitions

Let $E_j^{X_n}$ and its related matrices be given by

$$E_1^{X_1} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix}, \quad E_1^{X_2} = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad E_1^{X_3} = \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where we have used the e_1 basis for our example set. So, the basis set in its entirety is then defined as

$$\{E_j^{X_n}\} = \begin{pmatrix} E_1^{X_1} & E_1^{X_2} & E_1^{X_3} \\ E_2^{X_1} & E_2^{X_2} & E_2^{X_3} \\ E_3^{X_1} & E_3^{X_2} & E_3^{X_3} \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} & \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} & \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{pmatrix}.$$

The structure of these matrices reflects bivector properties familiar with geometric algebra and satisfy relations consistent with the $\mathfrak{su}(2)$ Lie algebra. These matrices form a 9-dimensional Lie algebra, as a subalgebra of $\mathfrak{gl}(3, \mathbb{C})$. The commutation relations for these matrices, as with the Pauli matrices, encode the algebraic rotation properties fundamental to the MegaCube geometry, collectively forming a closed algebra under matrix commutation.

2. Canonical In-Plane Commutators

The in-plane commutator relations match the $\mathfrak{su}(2)$ commutation structure up to scaling and complexification.

$$[E_1^{X_n}, E_2^{X_n}] = 2iE_3^{X_n} = \frac{1}{2}E_{12}^{X_n}, \quad [E_2^{X_n}, E_3^{X_n}] = -2E_1^{X_n} = \frac{1}{2}E_{23}^{X_n}, \quad [E_3^{X_n}, E_1^{X_n}] = -2iE_2^{X_n} = \frac{1}{2}E_{31}^{X_n}$$

3. Cross-Plane Commutators

The interactions between the X_1, X_2, X_3 planes yield nontrivial structure. The full set of generator commutation relations is shown in the table below, which behave as three interlinked $\mathfrak{su}(2)$ -like algebras in the X_1, X_2, X_3 planes. These are closed under mutual commutation, forming a higher-dimensional algebra similar to $\mathfrak{so}(6) \cong \mathfrak{su}(4)$.

$[E_j^{X_n}, E_k^{X_m}]$	$E_1^{X_1}$	$E_2^{X_1}$	$E_3^{X_1}$	$E_1^{X_2}$	$E_2^{X_2}$	$E_3^{X_2}$	$E_1^{X_3}$	$E_2^{X_3}$	$E_3^{X_3}$
$E_1^{X_1}$	0	$2E_{12}^{X_1}$	$2E_{13}^{X_1}$	$E_{32}^{X_1}$	$E_{31}^{X_1}$	$E_{31}^{X_1}$	$E_{23}^{X_1}$	$E_{13}^{X_1}$	$E_{31}^{X_1}$
$E_2^{X_1}$	$2E_{21}^{X_1}$	0	$2E_{23}^{X_1}$	$E_{31}^{X_1}$	$E_{23}^{X_1}$	$E_{32}^{X_1}$	$E_{13}^{X_1}$	$E_{32}^{X_1}$	$E_{32}^{X_1}$
$E_3^{X_1}$	$2E_{31}^{X_1}$	$2E_{32}^{X_1}$	0	$E_{13}^{X_1}$	$E_{23}^{X_1}$	0	$E_{13}^{X_1}$	$E_{23}^{X_1}$	0
$E_1^{X_2}$	$E_{23}^{X_1}$	$E_{13}^{X_1}$	$E_{32}^{X_1}$	0	$2E_{12}^{X_2}$	$2E_{13}^{X_2}$	$E_{31}^{X_1}$	$E_{31}^{X_1}$	$E_{32}^{X_1}$
$E_2^{X_2}$	$E_{13}^{X_1}$	$E_{32}^{X_1}$	$E_{32}^{X_1}$	$2E_{21}^{X_2}$	0	$2E_{23}^{X_2}$	$E_{31}^{X_1}$	$E_{23}^{X_1}$	$E_{32}^{X_1}$
$E_3^{X_2}$	$E_{13}^{X_1}$	$E_{23}^{X_1}$	0	$2E_{31}^{X_2}$	$2E_{32}^{X_2}$	0	$E_{31}^{X_1}$	$E_{23}^{X_1}$	0
$E_1^{X_3}$	$E_{32}^{X_1}$	$E_{31}^{X_1}$	$E_{31}^{X_1}$	$E_{23}^{X_1}$	$E_{13}^{X_1}$	$E_{31}^{X_1}$	0	$2E_{12}^{X_3}$	$2E_{13}^{X_3}$
$E_2^{X_3}$	$E_{31}^{X_1}$	$E_{23}^{X_1}$	$E_{32}^{X_1}$	$E_{13}^{X_1}$	$E_{32}^{X_1}$	$E_{32}^{X_1}$	$2E_{21}^{X_3}$	0	$2E_{23}^{X_3}$
$E_3^{X_3}$	$E_{13}^{X_1}$	$E_{23}^{X_1}$	0	$E_{13}^{X_1}$	$E_{23}^{X_1}$	0	$2E_{31}^{X_3}$	$2E_{32}^{X_3}$	0

Table 1. – Cross-Plane Commutator Relations

4. Cyclic Symmetry and Modular Rotations

The cyclic $\mathbb{Z}_3$ light generators $\mathcal{L}_a^+$ are defined as:

$$\mathcal{L}_0^+ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathcal{L}_1^+ = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathcal{L}_2^+ = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

of which, $\mathcal{L}_1^+, \mathcal{L}_2^+$ represent the primary generators of the $\mathbb{Z}_3$ group. Taking integer powers of the $\mathcal{L}_1^+$ generator, it is defined that

$$(\mathcal{L}_1^+)^n = \mathcal{L}_n^+, \quad n \equiv (\text{mod } 3), \quad \text{for } n \in \{0,1,2\}.$$

These cyclic operators act on the bivector generators to rotate the underlying algebraic plane:

$$\mathcal{L}_1^+ E^{X_1} \mathcal{L}_2^+ = E^{X_3}, \quad \mathcal{L}_1^+ E^{X_3} \mathcal{L}_2^+ = E^{X_2}, \quad \mathcal{L}_1^+ E^{X_2} \mathcal{L}_2^+ = E^{X_1}$$

This dual-sided action generalizes the notion of quaternionic or $SU(2)$ rotations, providing an elegant discrete model for moving between adjacent complex planes in the MegaCube geometry.

5. Vector Algebra and Hilbert Space Structure

To further analyze the MegaCube algebra, we map each generator matrix to a vector in $\mathbb{C}^3$ by summing over the rows:

$$\hat{e}_j^{X_n} = \sum_{\text{row}} E_j^{X_n}.$$

For instance, the $E_{ij}^{X_n}$ summations:

$$\begin{aligned} \hat{e}_{ij}^{X_1} &= \sum_{\text{row}} E_{ij}^{X_1} = \left\{ \begin{pmatrix} 0 \\ i \\ -i \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ i \\ i \end{pmatrix} \right\}, \\ \hat{e}_{ij}^{X_2} &= \sum_{\text{row}} E_{ij}^{X_2} = \left\{ \begin{pmatrix} -i \\ 0 \\ i \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} i \\ 0 \\ i \end{pmatrix} \right\}, \\ \hat{e}_{ij}^{X_3} &= \sum_{\text{row}} E_{ij}^{X_3} = \left\{ \begin{pmatrix} 0 \\ i \\ -i \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ i \\ i \end{pmatrix} \right\}. \end{aligned}$$

These vectors are elements of a Hilbert space $\mathcal{H} = \mathbb{C}^3$, equipped with the standard Hermitian inner product. The linear span of these vectors forms an invariant subspace under MegaCube algebra actions, much like the behavior of quantum states under $SU(2)$ rotations.

6. Modular Argand Transform and Visualization

To enable a visual representation of the $\{\hat{e}_j^{(X_n)}, \hat{e}_{ij}^{(X_n)}\}$ vector set $\in \mathbb{C}^3$, we perform the Argand plane modulo 3 transforms to each vector entry

$$\begin{pmatrix} \hat{e}_{(a)}^{X_1} \\ \hat{e}_{(b)}^{X_2} \\ \hat{e}_{(c)}^{X_3} \end{pmatrix} \rightarrow \begin{pmatrix} \text{Re}(\hat{e}_{(a)}^{X_1}) + \text{Im}(\hat{e}_{(c)}^{X_3}) \\ \text{Re}(\hat{e}_{(b)}^{X_2}) + \text{Im}(\hat{e}_{(a)}^{X_1}) \\ \text{Re}(\hat{e}_{(c)}^{X_3}) + \text{Im}(\hat{e}_{(b)}^{X_2}) \end{pmatrix} \in \mathcal{H}_{\mathbb{C}}^3,$$

where each vector entry $\hat{e}_{(\alpha)}^{X_n} \in \mathbb{C}$. In this form, we can define the transformed Argand plane modulo 3 vectors in the form $e_j(n, n+1), e_{ij}(n, n+1) \quad n \equiv (\text{mod } 3), n = \{1,2,3\}$.

IV. Modular Transform Geometry and Parametric Structure in the MegaCube

In **Figure 2**, we depict the modular geometry of the MegaCube, illustrating the action of Argand plane modulo-3 transforms on the vector structure arising from the geometric Lie algebra. This transform defines a cyclic mapping on the algebraic vector entries, aligning them with a modular plane tiling over $\mathbb{C}^3$. Overlaid on this structure is a distinguished black triangle, defined by three vertices that intersect specific points on the cubic MegaCube lattice.

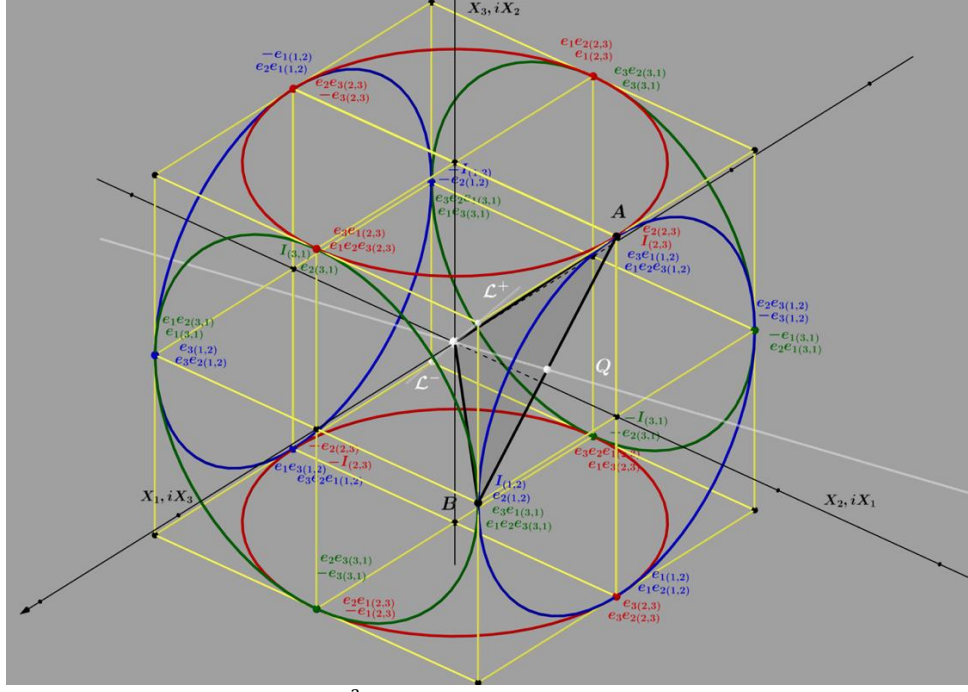


Figure 2. – Polyplex Lie Geometry image $\in \mathcal{H}_{\mathbb{C}}^3$, showing the arrangement of geometric algebra modular Argand vector elements $e_j(n, n+1), e_{ij}(n, n+1) \equiv (\text{mod } 3), n = \{1, 2, 3\}$, where $(n, n+1)$ indicates the rotation planes.

Let these lattice points be denoted A, B, O , with the segment $\overline{AB}$ supporting the charge axis q , which intersects the segment's midpoint. This axis corresponds to a direction in the Hilbert space $\mathcal{H}_{\mathbb{C}}^3$ where charge-related transformations are defined by the commutator algebra of matrix generators. The modular light generators $\mathcal{L}^+$ and $\mathcal{L}^-$, previously introduced as cyclic $\mathbb{Z}_3$ operators, act positively and negatively on the base algebra vectors, producing rotations between complex vector planes.

The segment $\overline{AB}$ is identified as a generator of a composite Lie algebra $\mathfrak{g} \subset \mathfrak{su}(3, \mathbb{C})$, combining rotation and scaling components from $\mathfrak{so}(3, \mathbb{C}) \oplus \mathfrak{u}(1)$. This algebra contains both Hermitian and anti-Hermitian operators and supports the inclusion of phase rotations and helicity basis shifts.

1. Elliptic Subgeometry in the Polyplex Lie Framework

To extract the geometric behavior along this segment, we consider the exponential map of a generator $A \in \mathfrak{g}$ acting on $\mathcal{H}_{\mathbb{C}}^3$:

$$U(\chi, \theta) = \exp((\chi, \theta)A)$$

Choosing

$$A = (I^{X_3} + E_2^{X_3}) + (E_{31}^{X_3} + E_{123}^{X_3}) = \begin{pmatrix} 1+i & 1+i & 0 \\ 1+i & 1+i & 0 \\ 0 & 0 & -\infty \end{pmatrix},$$

the matrix A is chosen to have an anisotropic spectrum that aligns with the desired elliptical deformation in the Argand-transformed Hilbert vector entries, such that the resulting flow follows quantized, elliptic orbits within a bivector-aligned complex plane. We compute:

$$U(\chi, \theta) = \exp \left[\begin{pmatrix} \chi + i\theta & \chi + i\theta & 0 \\ \chi + i\theta & \chi + i\theta & 0 \\ 0 & 0 & -\infty \end{pmatrix} \right] = \begin{pmatrix} \frac{1}{2}(\exp(\chi + i\theta) + 1) & \frac{1}{2}(\exp(\chi + i\theta) - 1) & 0 \\ \frac{1}{2}(\exp(\chi + i\theta) - 1) & \frac{1}{2}(\exp(\chi + i\theta) + 1) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Using the series expansion and properties of Lie algebra exponentiation, the above produces a two-parameter subgroup corresponding to elliptical rotation in the complexified bivector plane. This elliptic subgeometry does not exist in isolation but arises from the intersection of multiple components of the Polyplex structure. Specifically, when evaluating the modular Argand transform of the row-summed exponential image of a generator matrix $\exp((\chi, \theta)A)$, we obtain:

$$\hat{x}(\chi, \theta) = \text{Mod } 3 \left(\sum_{j=1}^n U(\chi, \theta)_{ij} \right)$$

This leads to an elliptic parametric equation for the transformed vector trajectory in $\mathbb{C}^3$, defined component-wise as:

$$\hat{x}(\chi, \theta) = \begin{pmatrix} \cos(\theta) \cosh(\chi) + i \sin(\theta) \sinh(\chi) \\ (\cos(\theta) + i \sin(\theta)) \exp(\chi) \\ \cos(\theta) \sinh(\chi) + i \sin(\theta) \cosh(\chi) \end{pmatrix}, \quad \theta \in [0, 2\pi), \quad \chi \in \mathbb{R}$$

The resulting curve lies on a complex ellipsoid, embedded within $\mathcal{H}_{\mathbb{C}}^3$, and intersects the MegaCube's cubic frame at rational or modular-aligned points. This elliptical structure inherits the algebraic properties of its generator and displays quantized phase rotations consistent with a discretized quantum spectrum. The interplay between the continuous elliptical motion, as generated by $U(\chi, \theta)$, and the discrete MegaCube lattice intersections reveals a striking geometric analogy to quantized orbits and quantum harmonic oscillators in curved internal symmetry spaces.

We consider $\hat{x}(\chi, \theta)$ over the reals, by defining $\hat{x}_{\mathbb{R}}(\chi, \theta)$ as

$$\hat{x}_{\mathbb{R}}(\chi, \theta) = \begin{pmatrix} \cos(\theta) \cosh(\chi) + \text{Im}(i \sin(\theta)) \sinh(\chi) \\ (\cos(\theta) + \text{Im}(i \sin(\theta))) \exp(\chi) \\ \cos(\theta) \sinh(\chi) + \text{Im}(i \sin(\theta)) \cosh(\chi) \end{pmatrix} = \begin{pmatrix} \cos(\theta) \cosh(\chi) + \sin(\theta) \sinh(\chi) \\ (\cos(\theta) + \sin(\theta)) \exp(\chi) \\ \cos(\theta) \sinh(\chi) + \sin(\theta) \cosh(\chi) \end{pmatrix}, \quad \theta \in [0, 2\pi), \quad \chi \in \mathbb{R},$$

which is appropriate, given that modular Argand transform represents overlapping real and imaginary trigonometric functions from distinct complex planes, and therefore, additive. This form demonstrates exponential asymmetry in the radial components, inducing eccentricity determined by the real parameter β . The eccentricity of $\hat{x}'(\chi, \theta)$ drops to zero when $\chi = -\ln(4/3)$, becoming a geodesic path on the singularity's surface:

$$\hat{x}_{\mathbb{R}}(-\ln(4/3), \theta) = \left(\frac{25}{24} \cos(\theta) - \frac{7}{24} \sin(\theta), \frac{3}{4} (\cos(\theta) + \sin(\theta)), -\frac{7}{24} \cos(\theta) + \frac{25}{24} \sin(\theta) \right)^T$$

As β increases, trajectory stretches elliptically, preserving orientation in a complexified plane anchored to the MegaCube's structural axes.

The $\hat{x}_{\mathbb{R}}(\chi, \theta)$ semi-major axes, occur at $\theta = \pm \pi/4$, and at $\pm 3\pi/4$ for the semi-minor axes. For a given value of $\chi \in \mathbb{R}$, the semi-major axis will have a modulus equal to

$$\|\hat{x}_{\mathbb{R}}(\chi, \pi/4)\| = \sqrt{\left(\frac{\cosh(\chi)}{\sqrt{2}} + \frac{\sinh(\chi)}{\sqrt{2}} \right)^2 + (\sqrt{2}e^{\chi})^2 + \left(\frac{\sinh(\chi)}{\sqrt{2}} + \frac{\cosh(\chi)}{\sqrt{2}} \right)^2} = \sqrt{3}e^{\chi},$$

and the modulus value for the semi-minor axis is

$$\|\hat{x}_{\mathbb{R}}(\chi, 3\pi/4)\| = \sqrt{\left(-\frac{\cosh(\chi)}{\sqrt{2}} + \frac{\sinh(\chi)}{\sqrt{2}} \right)^2 + \left(-\frac{\sinh(\chi)}{\sqrt{2}} + \frac{\cosh(\chi)}{\sqrt{2}} \right)^2} = e^{-\chi}.$$

Hyperbolae present in the plane of $\hat{x}(\chi, \theta)$ are defined by the following parametric equations:

$$\hat{\eta}(\beta)^+ = (\cosh(\ln(\beta)), e^{\ln(\beta)}, \sinh(\ln(\beta)))^T, \quad \beta \in \mathbb{R},$$

$$\hat{\eta}(\beta)^- = (i \sinh(\ln(\beta)), i e^{\ln(\beta)}, i \cosh(\ln(\beta)))^T, \quad \beta \in \mathbb{R}.$$

These hyperbolae are not arbitrary but result from elliptic intersections with the MegaCube and lattice geometries. These $\hat{x}(\chi, \theta)$ intersections occur at values of $\theta = \pi n/2, n \in \mathbb{Z}$. Mathematically, this is appreciated by evaluating $\hat{x}(\chi, \theta)$ for the values $\theta = 2\pi$ and $\pi/2$

$$\begin{aligned} \hat{\eta}(\beta)^+ &= \hat{x}(\chi, 2\pi) = \begin{pmatrix} \cos(2\pi) \cosh(\chi) + i \sin(2\pi) \sinh(\chi) \\ (\cos(2\pi) + i \sin(2\pi)) e^{\chi} \\ \cos(2\pi) \sinh(\chi) + i \sin(2\pi) \cosh(\chi) \end{pmatrix} = \begin{pmatrix} \cosh(\chi) \\ e^{\chi} \\ \sinh(\chi) \end{pmatrix}, \quad \chi \in \mathbb{R}, \\ \hat{\eta}(\beta)^- &= \hat{x}(\chi, \pi/2) = \begin{pmatrix} \cos(\pi/2) \cosh(\chi) + i \sin(\pi/2) \sinh(\chi) \\ (\cos(\pi/2) + i \sin(\pi/2)) e^{\chi} \\ \cos(\pi/2) \sinh(\chi) + i \sin(\pi/2) \cosh(\chi) \end{pmatrix} = \begin{pmatrix} i \sinh(\chi) \\ i e^{\chi} \\ i \cosh(\chi) \end{pmatrix}, \quad \chi \in \mathbb{R}. \end{aligned}$$

2. Curvature and Quantum Analogy

Using standard differential geometry, the curvature κ of this parametric ellipse embedded in $\mathbb{R}^3 \subset \mathbb{C}^3$ is calculated by:

$$\kappa(\chi, \theta) = \frac{|\hat{x}'_{\mathbb{R}}(\chi, \theta) \times \hat{x}''_{\mathbb{R}}(\chi, \theta)|}{|\hat{x}'_{\mathbb{R}}(\chi, \theta)|^3}$$

yielding a curvature function that modulates periodically in θ with local extrema corresponding to turning points on the ellipse. These turning points correspond to algebraically significant positions in the MegaCube lattice, where the rotated ellipsoid intersects discrete modular configurations.

This curvature analysis mirrors the quantized energy levels in a harmonic oscillator. Indeed, mapping these elliptical orbits to quantum state vectors allows us to define a discrete energy spectrum via the reduced Compton wavelength and assign thermal occupation probabilities through the Boltzmann distribution:

$$P_n = \frac{e^{-\beta E_n}}{Z}, \quad Z = \sum_n e^{-\beta E_n}$$

The energies $E_n \propto \hbar\omega(n + 1/2)$ align with the geometric oscillator model implicit in the ellipse, and the partition function is expressible as a Jacobi theta function, consistent with the elliptic modular symmetry inherent to the Polyplex framework.

3. Integration into the Polyplex Geometry

In the full geometry, these elliptical domains exist as submanifolds of the continuous component of Polyplex Lie Geometry, intersecting discretely with the MegaCube lattice. Their algebraic generators obey the closed commutator structure of the matrix Lie algebra underlying the framework, with basis vectors aligned to $\mathfrak{su}(3)$ and higher extensions. The elliptic section thus serves as both a local metric deformation and a harmonic quantization domain, capturing:

- Lie algebraic symmetry through bivector exponentiation,
- Modular Argand structure via discretized transforms,
- Quantum harmonic resonance through curvature and thermodynamics,
- Topological quantization through discrete MegaCube intersections.

This makes the elliptic structure a microcosm of Polyplex Lie Geometry's core: a unified, multi-metric, quantized algebraic space capable of encoding curvature, symmetry, quantum dynamics, and modularity simultaneously.

V. The Angle Space: A Framework for Geometric Manifolds and Lie Algebra Transformations

The concept of *angle space* provides an alternative foundation for analyzing geometric manifolds and their relationships, shifting the focus from Euclidean space to an angle-based representation. This framework is especially useful for understanding transformations between Lie algebras.

1. Core Concepts and Mathematical Definitions

Angle Space ($\in \mathbb{C}$): Instead of embedding geometric manifolds in a higher-dimensional Euclidean space $\in \mathbb{R}$, we consider them as open sets residing in a higher-dimensional *angle space* denoted $\in \mathbb{C}$. This is a foundational shift.

Exponential Mapping: This mapping plays a central role. It connects points on a complex unit circle to manifolds in complex space. If X is an "angle value," then e^{iX} maps points on the unit circle to specific geometric structures. We consider the exponential series for a single angle value X :

$$e^X = \sum_{n=0}^{\infty} \frac{X^n}{n!} = \frac{X^0}{0!} + \frac{X^1}{1!} + \frac{X^2}{2!} + \frac{X^3}{3!} + \frac{X^4}{4!} + \dots$$

This series can be separated into even (cosh) and odd (sinh) functions:

$$e^X = \left\{ 1 + \frac{X^2}{2!} + \frac{X^4}{4!} + \dots \right\} + \left\{ \frac{X^1}{1!} + \frac{X^3}{3!} + \frac{X^5}{5!} + \dots \right\} = \cosh(X) + \sinh(X)$$

For imaginary values iX , the exponential function is defined as:

$$e^{iX} = \left\{ 1 - \frac{X^2}{2!} + \frac{X^4}{4!} - \dots \right\} + \left\{ \frac{iX^1}{1!} - \frac{iX^3}{3!} + \frac{iX^5}{5!} - \dots \right\} = \cos(X) + i\sin(X)$$

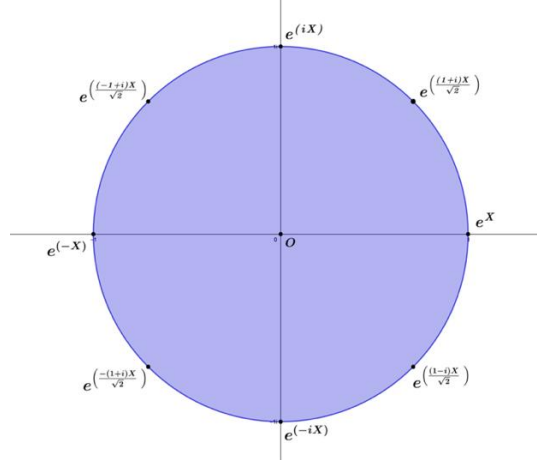


Figure 3. – The Angle Space Ring.

In the two-dimensional case, the complex angle space is best visualized as a complex unit circle embedded within the ambient space of angular variables. Through the action of the exponential map, each point on the boundary of this circle corresponds to a unique manifold defined by a complex-valued angle $\beta \in \mathbb{C}$. This construction gives rise to what we refer to as the *Angle Space Ring*, illustrated in **Figure 3.**, where each point on the boundary corresponds to a specific complex-angle parameter. These angular values, when exponentiated, yield unique manifolds governed by their respective tangent structures.

The ring, under exponential mapping, generates a distinct family of exponential functions, each encoding the transformation of a manifold via a Lie group representation parameterized by complex angles.

Definition – *Complex-Angle Space $\mathcal{B}_n$* : The Complex-Angle Space is defined as a complex n -dimensional angle space $\mathcal{B}_n$, which consists of all Lie subgroup representations parameterized by complex angular generators. Formally,

$$\mathcal{B}_n = \left\{ X^n(e^{i\beta_n}) \mid X^n \in \mathbb{R}^n, \beta_n \in \mathbb{C}, G = \exp(X^n(e^{i\beta_n})) \subset \text{Lie Group} \right\}$$

Each element $X^n(e^{i\beta_n})$ defines a tangent space to the group G , and all such tangent spaces lie on the boundary $\partial\mathcal{B}_n$. Thus, the boundary of the complex-angle space serves as the holographic shell upon which all infinitesimal generators of the group's exponential geometry are encoded.

This formulation allows the complex-angle space $\mathcal{B}_n$ to serve as the geometric domain for angular parameterization of Lie groups and their associated tangent structures. It is within this boundary that angular morphisms, curvature variations, and projectional structures of Polyplex Lie Geometry are realized.

VI. Lie groups and Their Transformations

Definition – Let $G(X), G(Y) \subset \mathcal{B}$ be arbitrary Lie groups, representing closed subsets of the Complex-Angle Space $\mathcal{B}_n$. $\Omega: G(X) \rightarrow G(Y)$ is a bijective and continuous mapping that takes the Lie group $G(X^n) \in \mathcal{B}_n$ to $G(Y^n) \in \mathcal{B}_n$,

$$\Omega: G(X^n) \rightarrow G(Y^n) \in B(n, \mathbb{C}).$$

We define the function mapping $\Omega: G(X^n)$, for a chosen complex angle transformation $e^{i\beta_n}$

$$\Omega: G(X^n) = G(X^n e^{i\beta_n}),$$

for $X^n e^{i\beta_n}$, representing the element-to-element multiplicative product of the two $n \times n$ matrices.

For the $n = 1$ dimensional case, the mapping $\Omega: G(X^n) \rightarrow G(Y^n)$ is straightforward, appreciated for three-dimensional mapping example

$$\Omega: G(X_1^{3\mathbb{C}}) \rightarrow G(X_1^{3\mathbb{R}}),$$

for $G(X_1^{3\mathbb{C}})$, a generator from the $D_{3\mathbb{C}}$ dihedral group, and $G(X_1^{3\mathbb{R}})$, a generator from the $D_{3\mathbb{R}}$ group. We set $\beta = -\pi/2$, while recalling the function $X \circ \beta$ represents the resulting matrix formed by the element-to-element multiplication products $(x_{j,k} \cdot \beta_{j,k})$, finding

$$\Omega: G(X^3) = G(X^3 \circ e^{i\beta_3}) = \exp \left(\begin{pmatrix} ix_1 & 0 & 0 \\ 0 & 0 & ix_1 \\ 0 & ix_1 & 0 \end{pmatrix} \circ \begin{pmatrix} e^{-\frac{i\pi}{2}} & 0 & 0 \\ 0 & 0 & e^{-\frac{i\pi}{2}} \\ 0 & e^{-\frac{i\pi}{2}} & 0 \end{pmatrix} \right) = \exp(X_1^{3\mathbb{R}}) = G(X_1^{3\mathbb{R}}).$$

The dihedral forms of the Lie groups that we have been working with belong to the general linear group $GL(3, \mathbb{C}) \in \mathbb{C}^3$

$$G(X^3 \circ e^{i\beta_3}) \in GL(3, \mathbb{C}).$$

The $\Omega(X)$ mapping is bijective, and given that these groups are all continuous, the topological spaces $G(X^n), G(Y^n)$ are homeomorphic to one another.

Theorem – For any two topological spaces of the form $G(X), G(Y)$, located on the boundary ∂B in complex-angle space, there is a homomorphism between $G(X)$ and $G(Y)$.

Proof – The mapping $\Omega: G(X) \rightarrow G(Y)$ of any two functions of the form $G(X), G(Y)$ located on the boundary ∂B in complex-angled space is a bijective (one-to-one) and continuous mapping, and there is an inverse mapping $\Omega^{-1}: G(Y) \rightarrow G(X)$ for any mapping $\Omega: G(X) \rightarrow G(Y)$, that is also bijective and continuous. The mapping,

$$\Omega: G(X^n) = G(X^n \circ e^{i\beta_n}) = G(Y^n),$$

maps each element of X^n to Y^n bijectively. For every mapping $G(X^n \circ e^{i\beta_n})$ there is an inverse mapping,

$$\Omega^{-1}: G(Y^n) = G(Y^n \circ e^{i(-\beta)_n}) = G(X^n),$$

by way of complex conjugation $\overline{e^{i\beta_n}} = e^{i(-\beta)_n}$, which maps $G(Y^n)$ back to $G(X^n)$.

□

To express the mathematical framework more intuitively, we consider the action of angle-based transformations in complex space, where one manifold is rotated into another via a well-defined map in angle space. Specifically, for the Lie group structures arising from the dihedral form, these can be embedded and visualized on the surface of a complex-angle sphere, a three-dimensional generalization of the two-dimensional projection illustrated in **Figure 4**.

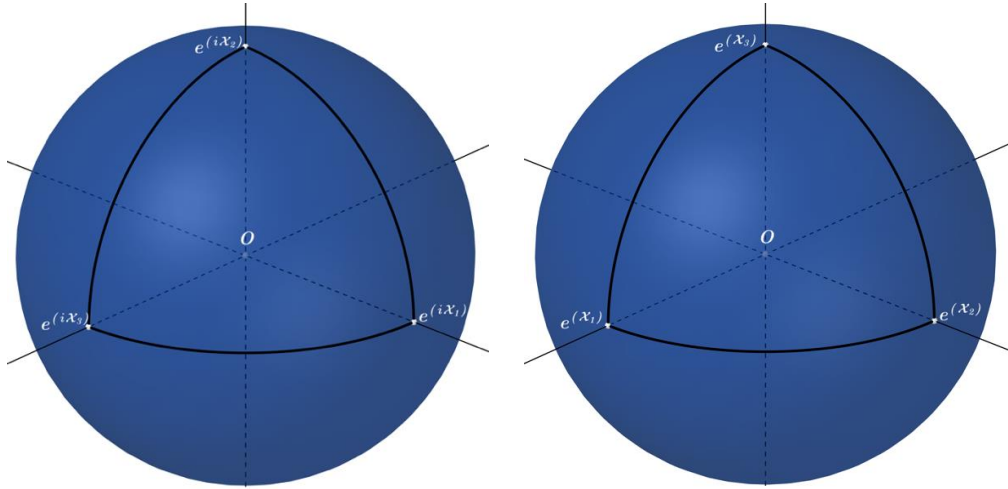


Figure 4.- Angle Space Transformation $\Omega: G^{3\mathbb{C}}(X_1 + X_2 + X_3)$ (left) $\rightarrow G^{3\mathbb{R}}(X_1 + X_2 + X_3)$ (right)

Taking the dihedral group $D_3 \in \mathbb{C}^{3 \times 3}$ as an example, it can be mapped into a hyperbolic Lie form by applying the transformation:

$$\Omega: D^{3\mathbb{C}}(X_1 + X_2 + X_3) \rightarrow D^{3\mathbb{R}}(X_1 + X_2 + X_3)$$

That is, by setting $\beta_1, \beta_2, \beta_3 = -\pi/2$, we can evaluate the corresponding algebraic mapping between the group representations:

$$\Omega: D^{3\mathbb{C}}(X_1 + X_2 + X_3) = D^{3\mathbb{C}}(X_1 \circ e^{i\beta_1}, X_2 \circ e^{i\beta_2}, X_3 \circ e^{i\beta_3}) = D^{3\mathbb{R}}(X_1 + X_2 + X_3).$$

VII. Extension to Tangent Bundles

When $\mathcal{B}_n$ is considered in the context of special linear and special unitary groups, its structure becomes richer and deeply tied to complex geometry.

For $SL(n, \mathbb{C})$, which consists of complex $n \times n$ matrices with determinant 1, the Lie algebra $\mathfrak{sl}(n, \mathbb{C})$ is defined as:

$$\mathfrak{sl}(n, \mathbb{C}) = \{A \in \mathbb{C}^{n \times n} | \text{Tr}(A) = 0\}$$

In this context, each element $X^n(e^{i\beta_n}) \in \mathfrak{sl}(n, \mathbb{C})$ defines a fiber over a point in the base complex manifold, and the total space of these tangent vectors defines the tangent bundle $TSL(n, \mathbb{C})$. The complex-angle space $\mathcal{B}_n$ parametrizes the local angular structure of each fiber direction in the bundle. The boundary $\partial\mathcal{B}_n$ serves as a domain for defining connection 1-forms A and curvature 2-forms $F = dA + A \wedge A$, describing how these tangent spaces twist globally across the manifold.

For $SU(n)$, the special unitary group, whose elements preserve Hermitian inner product and have unit determinant, the tangent space at identity is given by the algebra:

$$\mathfrak{su}(n) = \{A \in \mathbb{C}^{n \times n} | A^\dagger = -A, \text{Tr}(A) = 0\}$$

Each tangent vector $X^n(e^{i\beta_n}) \in \mathfrak{su}(n)$ describes a skew-Hermitian transformation that, when exponentiated, maps into $SU(n)$. In this context, $\mathcal{B}_n$ becomes a parameter space for Hermitian phase rotations, encoding directions of infinitesimal motion on the compact Lie group manifold. These rotations form the local trivializations of the principal $SU(n)$ -bundle, and the complex-angle values define the transition functions between fibers.

The geometric content of $\mathcal{B}_n$ in this setting enables a Clifford-algebraic embedding of unitary symmetry in the Polyplex framework. The angular variables $\beta_n \in \mathbb{C}$ correspond to spinorial rotations and local gauge transformations, and the global structure of the fiber bundle encodes holonomy, curvature, and topological charge (e.g., Chern classes).

Thus, the complex-angle space $\mathcal{B}_n$ serves as a geometric and algebraic scaffold for defining tangent spaces, connection structures, and gauge fields in the context of high-dimensional symmetry groups:

- In $SL(n, \mathbb{C})$, it governs holomorphic structure and volume-preserving flows.
- In $SU(n)$, it encodes unitary symmetries, gauge curvature, and phase dynamics.
- In both, the boundary $\partial\mathcal{B}_n$ forms a fiber-generating shell, mapping infinitesimal angular transformations to physical or geometric projections via the exponential map.

This formalism plays a foundational role in Polyplex Lie Geometry, where the infinite sphere of tangent spaces and its Angle Space mediation provide a universal model of projection from Unity into structured multiplicity, uniting the differential, algebraic, and metaphysical domains into a single coherent geometry.

VIII. Vector Representations in Complex-Angled Space

The polar visualization of the complex unit circle demonstrates that a radial vector may be interpreted as a combination of an imaginary-valued sine function and a real-valued cosine function, aligned along the vertical and horizontal axes, respectively. This interpretation extends naturally to the geometry of dihedral Lie groups, where radial vector values are used to represent distinct group elements on the boundary $\partial\mathcal{B}_n$ of the complex-angle space. In this picture, the imaginary component arises from the exponential function projected vertically, while the real component contributes along the horizontal axis **Figure 3**.

Let the vector $\vec{R}$ represent the real-valued exponential function $e^{r \cos(\theta)}$, and $\vec{I}$ represent the imaginary exponential function $e^{i \sin(\theta)}$. Their product vector $\vec{E} = \vec{R} \cdot \vec{I}$ forms the full exponential map $\exp(re^{i\theta})$, which lies on the boundary of the complex unit disc and represents a point on the complex-angle sphere.

This two-dimensional construction holds for dihedral Lie group generators. In a similar way, for a group embedded in higher-dimensional space such as $\mathcal{B}_n$, we find the product of two exponential function vectors:

$$\vec{E} = e^{X^n(e^{i\beta_n})} = e^{X^n \cos(\beta_n)} \cdot e^{iX^n \sin(\beta_n)}$$

The associated matrix form of these vectors corresponds to:

$$G = \exp\left(X^n(e^{i\beta_n})\right) = \exp(X^n \cos(\beta_n)) \cdot \exp(iX^n \sin(\beta_n))$$

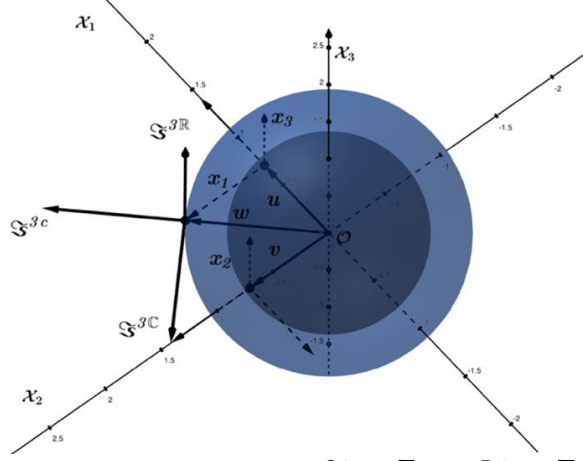


Figure 5. – Tangent Space Vector Addition: $\mathcal{T}^{3\mathbb{C}}(X_1/\sqrt{2}) + \mathcal{T}^{3\mathbb{R}}(X_1/\sqrt{2}) = \mathcal{T}^{3\mathbb{C}}(X_1)$

To examine the corresponding tangent vectors, we take the matrix logarithm of the group elements, yielding the infinitesimal generators in the Lie algebra:

$$\ln(G) = X^n(e^{i\beta_n})$$

The tangent space on the boundary $\partial\mathcal{B}_n$ is then constructed through vector addition of these infinitesimal elements. That is,

$$\mathcal{T} = \ln(e^{X_1}) + \ln(e^{X_2}) = X_1 + X_2$$

This additive property allows us to compose the tangent space vectors geometrically. Expanding the addition gives:

$$\mathcal{T} = X_1^n(e^{i\beta_1^n}) + X_2^n(e^{i\beta_2^n})$$

Figure 5. illustrates this result, where two tangent vectors $\mathcal{T}_1$ and $\mathcal{T}_2$, originating at distinct points on the complex-angle sphere of unit radius, are added vectorially. The resulting tangent vector $\mathcal{T}$ lies on the boundary $\partial\mathcal{B}_n$. Each component vector remains oriented according to its angular embedding:

- The vector $\vec{E}$, representing the full exponential form, remains confined to the geodesic plane of the sphere.
- The vectors $\vec{R}$ and $\vec{I}$ align along their respective real and imaginary axes, maintaining orthogonality with respect to their angular base.

These orientations reflect the internal structure of the dihedral group D_n , where reflections and rotations correspond to angular permutations. In particular, the generator element $r \in D_n$, which maps points $\theta \mapsto \theta + 2\pi/n$, maintains the geometric integrity of tangent vector orientations as they evolve along the complex-angle boundary.

Thus, this vector-based construction provides a concrete realization of the Lie algebra as a directional structure within the angle space, where each exponential and logarithmic operation reflects a precise movement or deformation of the tangent field around the singularity. The geometry of $\mathcal{B}_n$ and its boundary $\partial\mathcal{B}_n$ therefore not only encodes the infinitesimal symmetries of Lie group structure but also provides a visual and vectorial intuition for how complex-angled transformations propagate through higher-order Polyplex Lie geometry.

IX. The Infinite Sphere of Tangent Spaces: A Holographic Boundary of Lie Algebraic Symmetry

The Infinite Sphere of Tangent Spaces denoted $\mathcal{T}_\infty$, constitutes a central geometric structure in Polyplex Lie Geometry. It forms the complex boundary of the singularity $\mathcal{S}_0$, encoding the complete symmetry content of the infinite-dimensional Lie group $\mathcal{G}_\infty$. In this construction, the tangent spaces are not merely local linear approximations, but rather the infinitesimal generators of all group-based geometry. They manifest as boundary-valued Lie algebras: $\mathfrak{g} \subset \partial\mathcal{T}_\infty$, with each $\mathfrak{g}_\beta = X(e^{i\beta})$ corresponding to a unique complex angle in $\mathcal{B}_n$.

1. Dual Exponential Structure and Polar Decomposition

Each tangent space element arises via exponential mapping from a subtangent form:

$$G = \exp(\exp(\mathfrak{g}_\beta)), \text{ where } \mathfrak{g}_\beta = X(e^{i\beta})$$

This nested exponential reflects a projection from the singular kernel $\mathcal{S}_0$ through the boundary $\partial\mathcal{T}_\infty$ and outward into the extended Polyplex manifold. The corresponding subtangent space $\mathfrak{g}_\beta$ is obtained via a double logarithmic pullback:

$$\mathfrak{g}_\beta = \ln(\ln(G))$$

Expressed in polar coordinates, each element of $\mathfrak{g}_\beta$ admits decomposition into radial modulus r and angular parameter β :

$$\mathfrak{g}_\beta = r \cdot X(e^{i\beta}), \text{ with } r \in \mathbb{R}^+, \beta \in [0, 2\pi)$$

This dual nature of angle and radius becomes foundational for curvature, dilation, and quantization within the Polyplex structure. As $r \rightarrow 1$, the exponential image remains confined to $\partial\mathcal{T}_\infty$, yielding unitary transformations in the tangent group $\mathcal{G}_\mathcal{T}$. When $r \neq 1$, the mapping transitions to extended or contracted geometries corresponding to dilation (or compression) away from the singularity.

2. Matrix Structures and Hyperbolic-Circular Decomposition

To express the full analytic structure of the tangent group $G_\mathcal{T}$, we employ the polar and hyperbolic decomposition of the generating matrices:

$$G_\mathcal{T} = \begin{pmatrix} \cosh(r) & i\sinh(r) \\ i\sinh(r) & \cosh(r) \end{pmatrix} \times \begin{pmatrix} \cos(\beta) & -\sin(\beta) \\ \sin(\beta) & \cos(\beta) \end{pmatrix}$$

Here, the hyperbolic matrix captures the radial dilation, while the circular matrix encodes angular rotations along $\mathcal{B}_n$. This structure enables interpretation of tangent group flows as simultaneous hyperbolic-spherical transformations in the embedded Lie group manifold. Notably, the determinant of such composite matrices becomes:

$$\det(G_\mathcal{T}) = \cosh^2(r) - \sinh^2(r)$$

preserving unitarity when mapped on the boundary and enabling volume-preserving flows when extended.

3. Waveform Behavior and Boundary Oscillations

By evaluating each matrix entry in terms of (r, β) , we obtain real and imaginary waveforms that exhibit periodicity on $\partial\mathcal{T}_\infty$:

- Real-valued maxima: $\beta = 2\pi n$
- Imaginary-valued maxima: $\beta = (2n + 1)\pi$

The identity elements of the Lie algebra appear at these turning points. The frequency of the imaginary waveform (governed by sine) is twice that of the real waveform (governed by cosine), reflecting an intrinsic double-cover behavior of the angle space.

Hyperbolic waveform forms dominate in the r -domain, while oscillatory circular forms dominate in the β -domain, yielding the full hybrid geometry of $\mathcal{T}_\infty$.

4. Angle Space Rotations and Subalgebra Embeddings

The angle space $\mathcal{B}_n$ serves as a rotational shell over $\mathcal{T}_\infty$, where transformations of the form:

$$\mathfrak{g}_\beta + \theta = R(\theta) \cdot \mathfrak{g}_\beta \cdot R^{-1}(\theta)$$

describe local fiber rotations in the tangent bundle. Here, $R(\theta)$ acts as an angle-preserving automorphism in the fiber group, analogous to phase rotations in a $U(1)$ bundle. These rotations preserve curvature but alter directionality, embedding a local spinor geometry in the tangent sphere.

Further, direct product embeddings such as:

$$\mathfrak{g}_{SU(n)} \oplus \mathfrak{g}_{SL(n, \mathbb{C})} \subset \partial\mathcal{T}_\infty$$

are realized through alignment of radial dilations (SL) with unitary phase rotations (SU), establishing tangent bundles with both conformal and gauge symmetries.

5. The Tangent Space Complex Symmetry Group

We now define the total symmetry group of the tangent boundary:

Definition – The Tangent Space Complex Symmetry Group $\mathcal{G}_\mathcal{T}$ is the set of functions under composition, formed by unitary exponentials of subtangent Lie algebras with both radial and angular components:

$$\mathcal{G}_\mathcal{T} := \{ \exp(rXe^{(i\beta)}) \mid r \in \mathbb{R}^+, \beta \in [0, 2\pi) \}$$

This group is closed under composition, associative, possesses an identity (at $r = 0$) and each element admits an inverse under angular conjugation. It spans all holomorphic and anti-holomorphic symmetry directions over the boundary $\partial\mathcal{T}_\infty$.

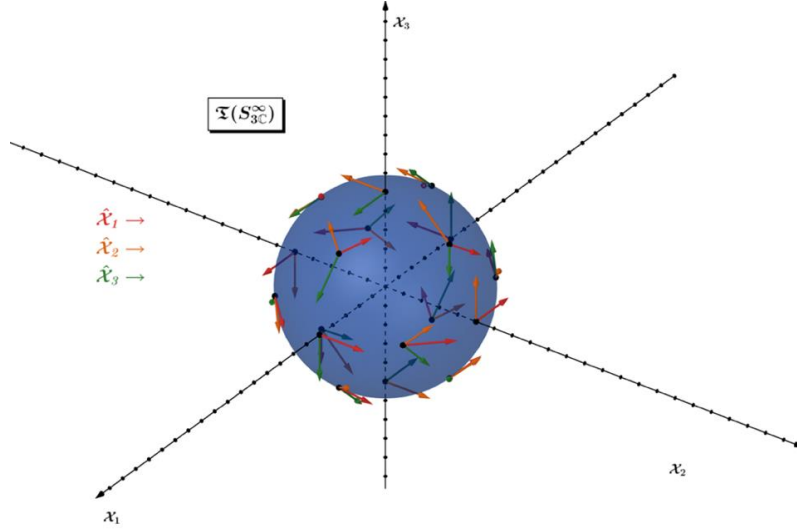


Figure 6. – The Complex Tangent Space Symmetry Group as a subgroup of the infinite number of tangent spaces on the ISTS.

The angle space and the *Infinite Sphere of Tangent Spaces (ISTS)* are deeply intertwined in the geometric and algebraic architecture of Polyplex Lie Geometry. Rotations within the angle space occur through modulations of the complex phase parameter β_n , embedded in the Lie group formulation $G(\chi e^{i\beta})$. These phase variations correspond to continuous transformations within the Lie algebra, specifically, to motions across the tangent space defined at the boundary of the complex-angle sphere. However, underlying these transformations is a more fundamental layer: the subtangent space. From the perspective of Lie theory, the subtangent space acts as a causal pre-structure, its elements are mapped exponentially to generate the tangent vectors themselves. This hierarchical relation permits smoothly parameterized rotations in the tangent space, which are geometrically interpreted as deformations or transitions between distinct metric geometries within the Polyplex framework. Because these transitions are driven by nested exponentials (exp of log of exp), movement through the subtangent space not only orchestrates angular permutations but also enables curvature-altering transformations, allowing Polyplex Lie Geometry to fluidly interlink spherical, hyperbolic, and other geometric regimes via algebraically continuous flows across ISTS.

X. The Elliptic Group

Within the Polyplex Lie Geometry framework, the elliptic group $\mathcal{E}_\eta(\theta, \chi)$ emerges as a central analytic structure representing a light-algebra symmetry subgroup. This group encapsulates the geometric behavior of elliptical deformation and oscillation within the continuous metric manifold, and its intersectional properties with hyperbolic curves and light axes define critical transition zones between metric geometries. The elliptic group thereby bridges curvature-dense domains and tangent-shell symmetries, linking curved algebraic motion with quantized boundary intersections in the modular MegaCube lattice. We begin by defining $\mathcal{E}_\eta(\theta, \chi)$ in its group form:

$$\mathcal{E}_\eta(\theta, \chi) = \begin{pmatrix} \frac{1}{2}(e^{(i\theta+\chi)} + 1) & \frac{1}{2}(e^{(i\theta+\chi)} - 1) \\ \frac{1}{2}(e^{(i\theta+\chi)} - 1) & \frac{1}{2}(e^{(i\theta+\chi)} + 1) \end{pmatrix}$$

This matrix defines a generalized elliptical transformation in a bivector-aligned basis, embedding phase-rotated shear dynamics into the geometry. For static χ values, the determinant of the transformation confirms its preservation of area within the transformation plane:

$$|\mathcal{E}_\eta(\theta, \chi)| = \frac{1}{4}(e^{(i\theta+\chi)} + 1)^2 - \frac{1}{4}(e^{(i\theta+\chi)} - 1)^2 = e^{i\theta+\chi}$$

The parametric form of the group is defined:

$$\mathcal{E}_\eta(\theta, \chi) = (\cosh(i\theta + \chi), e^{i\theta+\chi}, \sinh(i\theta + \chi))$$

This θ -dependent parametric group evolves elliptical motion with separate modulation along the χ and θ axes, encoding anisotropy into the group evolution. The angle variables χ and θ represent separate rapidity/angular velocity scales in hyperboloid and spherical geometries, respectively, introducing asymmetry from the light axes, critical for generating eccentricity in embedded orbits. This form is interpreted as a Lie group trajectory whose generator matrix contains diagonal and off-diagonal bivector content with elliptical phase offset.

Elliptic group dynamics naturally intersect with hyperbolic geometry in Polyplex Lie Geometry, forming a correspondence between curved flows and saddle-point divergence. Ellipses will intersect each hyperbolic curve for the following values of θ :

$$\mathcal{E}_\eta(\theta, \chi) \cap \eta_{(+, \chi)} \text{ for } \theta = 2\pi k, \quad k \in \mathbb{Z},$$

$$\mathcal{E}_\eta(\theta, \chi) \cap \eta_{(-, -\chi)} \text{ for } \theta = 2\pi k + \pi/2, \quad k \in \mathbb{Z},$$

$$\mathcal{E}_\eta(\theta, \chi) \cap -\eta_{(+, \chi)} \text{ for } \theta = 2\pi k + \pi, \quad k \in \mathbb{Z},$$

$$\mathcal{E}_\eta(\theta, \chi) \cap -\eta_{(-, -\chi)} \text{ for } \theta = 2\pi k - \pi/2, \quad k \in \mathbb{Z}$$

For the two hyperbolae along the positive light axis (charge axis Q^+), the distance between elliptic-to-hyperbola intersections is found by the following:

$$\begin{aligned} & (\cosh(2i\pi + \chi), e^{2i\pi + \chi}, \sinh(2i\pi + \chi)) - (\cosh(i\pi/2 + \chi), e^{i\pi/2 + \chi}, \sinh(i\pi/2 + \chi)) \\ &= (-\cosh(\chi) + i\sinh(\chi), (1-i)e^\chi, \sinh(\chi) - icosh(\chi)) \\ &\rightarrow (-\cosh(\chi) + |i\sinh(\chi)|, e^\chi - |ie^\chi|, \sinh(\chi) - |icosh(\chi)|) \\ &\rightarrow \|\cosh(\chi) + |i\sinh(\chi)|, e^\chi - |ie^\chi|, \sinh(\chi) - |icosh(\chi)|\| = \sqrt{2}|e^{-\chi}| \end{aligned}$$

This formula demonstrates that the separation distance between elliptic-hyperbolic crossings is governed by the square root of the angular asymmetry between the χ and θ rotation rates. When $\chi = -\ln(3)/4$, the system approaches the singularity state, recovering circular rather than elliptical motion. Thus, angular asymmetry governs both the eccentricity of the elliptical form and the frequency of its intersection with the hyperbolic shell. For $\chi = -\ln(3)/4$, we define the geodesic parametric equation on the singularity, as

$$G_\eta(\theta) = \begin{pmatrix} \cos(\theta) \cosh\left(\frac{\ln(3)}{4}\right) - \sin(\theta) \sinh\left(\frac{\ln(3)}{4}\right) \\ (\cos(\theta) + \sin(\theta))e^{-\left(\frac{\ln(3)}{4}\right)} \\ -\cos(\theta) \sinh\left(\frac{\ln(3)}{4}\right) + \sin(\theta) \cosh\left(\frac{\ln(3)}{4}\right) \end{pmatrix}$$

In the Polyplex Lie Geometry context, $\mathcal{E}_\eta(\theta, \chi)$ serves as the localized symmetry generator for elliptical subspaces, domains where phase-rotated bivectors and harmonic deformation dominate the geometry. These elliptical regions arise naturally within the modular framework as submanifolds of the continuous metric component, where real and imaginary components of vector flow form bounded orbits within a rotationally sheared complex plane. As the group evolves through its parameter space, intersections with hyperbolic domains mark critical curvature transitions, indicating quantum phase change, geometric resonance, or eigenvalue crossings of the underlying operator algebra.

The elliptic group section thus formalizes a critical symmetry regime within Polyplex Lie Geometry, one that both modulates the curvature spectrum of the continuous domain and synchronizes with the modular intersections of the MegaCube lattice. The group $\mathcal{E}_\eta(\theta, \chi)$ enables anisotropic transformations, elliptical encodings of Lie algebraic flow, and defines metric thresholds at which geometries transition between bounded (elliptic), unbounded (hyperbolic), and flat (parabolic) substructures. It anchors the geometric dynamics to the algebraic roots of angular deformation, spectral splitting, and quantized phase motion. As such, it completes the harmonic triad of spherical, hyperbolic, and elliptic structures embedded within the Polyplex continuum.

To visualize the elliptic group $\mathcal{E}_\eta(\theta, \chi)$ in the Polyplex Lie Geometry framework, the elliptic group $\mathcal{E}_\eta(\theta, \chi)$, we set $\chi = \pm 1$, defining two elliptic curves:

$$\mathcal{E}_\eta(\theta, 1) = (\cos(\theta) \cosh(1) + \sin(\theta) \sinh(1), (\cos(\theta) + \sin(\theta))e^1, \cos(\theta) \sinh(\theta) + \sin(\theta) \cosh(\theta))^T$$

$$\mathcal{E}_\eta(\theta, -1) = (\cos(\theta) \cosh(1) - \sin(\theta) \sinh(1), (\cos(\theta) + \sin(\theta))e^{-1}, -\cos(\theta) \sinh(\theta) + \sin(\theta) \cosh(\theta))^T,$$

Hyperbolic equations η are defined:

$$\eta_{(+, +)} = (\cosh(\chi), e^\chi, \sinh(\chi))^T,$$

$$\eta_{(-,-)} = i(\sinh(\chi), e^\chi, \cosh(\chi))^T$$

In **Figure 7**, we plot the above equations within the *MegaCube* lattice, illustrating their geometric embedding alongside the geodesic equation $G_\eta(\theta)$. The charge axis denoted Q^+ , marks the light-algebraic direction asymptotic to the hyperbolic-orthogonal curves $\eta_{(+,+)} , \eta_{(-,-)}$. Orthogonal to this axis, the elliptic curves $\mathcal{E}_\eta(\theta, 1)$ and $\mathcal{E}_\eta(\theta, -1)$ are oriented precisely within the plane defined by the geodesic $G_\eta(\theta)$, forming a conjugate geometric system. This alignment underscores the curvature symmetry between hyperbolic divergence and elliptic boundedness in the Polyplex framework, while also highlighting the orthogonal structure inherent to the *MegaCube*'s modular topology.

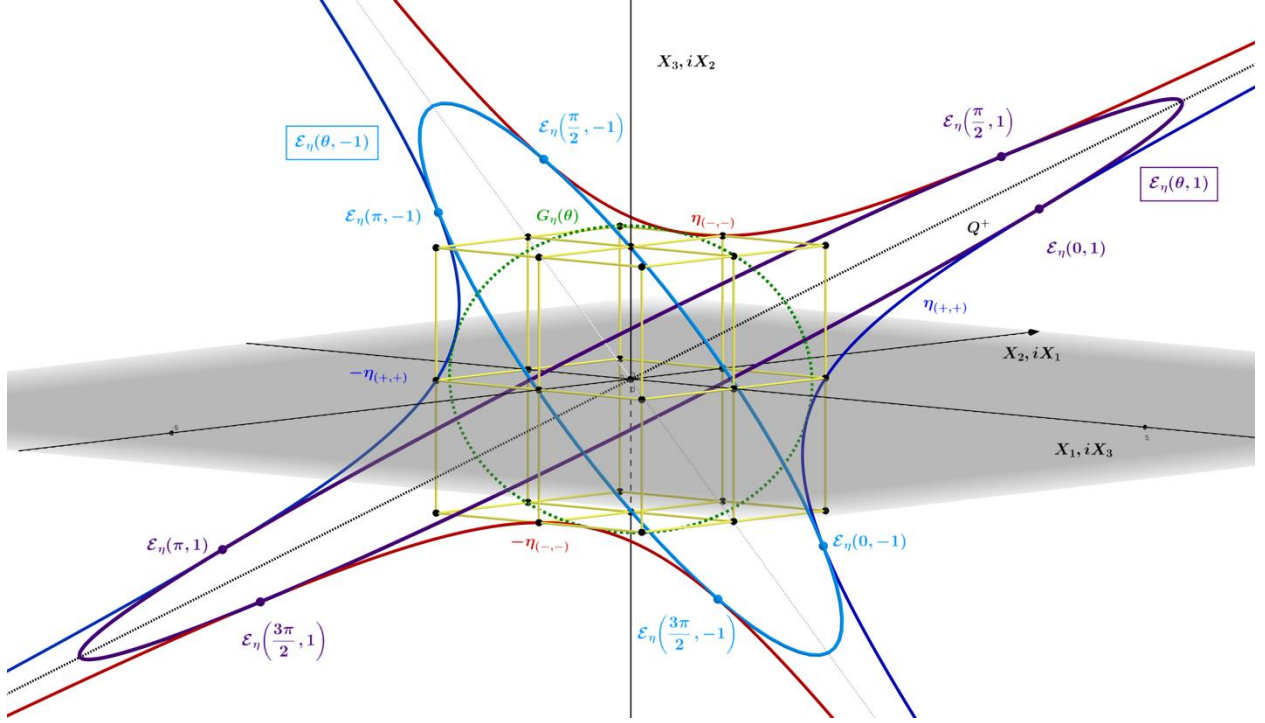


Figure 7. – Elliptic $\mathcal{E}_\eta(\theta, \pm 1)$ and hyperbolic $\pm\eta_{(\pm,\pm)}$ curves relationship to the *MegaCube* lattice, the $G_\eta(\theta)$ geodesic curve, and the light axis in Polyplex Lie Geometry.

XI. The Third Observer – The Light Glowney-Dirac Delta

The Third Observer, denoted through the Light Glowney-Dirac Delta formalism, serves as a localized, quantized reference frame within the Polyplex Lie Geometry framework. It functions as a geometric probe embedded at the boundary between curved continuum and discrete modularity, offering a structural anchor for symmetry-breaking and observational locality.

Unit vectors on the circle for the light delta can be defined as

$$v_1 = \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right), \quad v_2 = \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right).$$

The circle's centroid C_δ equals

$$C_\delta = \frac{1}{2}(\cosh(\chi), e^\chi, \sinh(\chi)) + \frac{1}{2}(\sinh(\chi), e^\chi, \cosh(\chi)) = \left(\frac{e^\chi}{2}, e^\chi, \frac{e^\chi}{2}\right),$$

which will be a distance:

$$\Delta = \|C_\delta\| = \sqrt{\left(\frac{e^\chi}{2}\right)^2 + (e^\chi)^2 + \left(\frac{e^\chi}{2}\right)^2} = \frac{\sqrt{3}}{\sqrt{2}}e^\chi$$

from the origin of the MegaCube. This fixed displacement defines a symmetric anchor point from which the radial symmetry of the trumpet curvature is observed. The offset is not arbitrary: it reflects a stable, phase-shifted centroid embedded within the Hilbertian field that underlies Polyplex geometry.

For any value $\chi \in \mathbb{R}$, the circle's radius will equal

$$\begin{aligned} R &= \left\| \frac{1}{2}(\cosh(\chi), e^\chi, \sinh(\chi)) - \frac{1}{2}(\sinh(\chi), e^\chi, \cosh(\chi)) \right\| \\ &= \left\| \left(\frac{e^{-\chi}}{2}, 0, -\frac{e^{-\chi}}{2} \right) \right\| = \sqrt{\left(\frac{e^{-\chi}}{2} \right)^2 + \left(-\frac{e^{-\chi}}{2} \right)^2} = \frac{e^{-\chi}}{\sqrt{2}}. \end{aligned}$$

The equation of the circle centered at C_δ , representing a path orthogonal to the curvature of Torricelli's trumpet, is defined as:

$$S_p = C_\delta + R(\cos(\theta) v_1 + \sin(\theta) v_2), \quad t \in \mathbb{R}$$

Using our derived radius and center point functions, for any $\chi \in \mathbb{R}$, the equation of the circle becomes

$$S(\theta, \chi) = \left(\frac{e^\chi}{2}, e^\chi, \frac{e^\chi}{2} \right) + \frac{e^{-\chi}}{\sqrt{2}} \left(\cos(\theta) \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right) + \sin(\theta) \left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \right)$$

This leads to the parametric trumpet equation, where the evolving radius and rotating center trace out a surface of revolution corresponding to the inverse hyperboloid domain:

$$S(\theta, \chi) = \left(\frac{e^\chi}{2} + e^{-\chi} \left(-\frac{\cos(\theta)}{2} + \frac{\sin(\theta)}{\sqrt{6}} \right), e^\chi - e^{-\chi} \left(\frac{\sin(\theta)}{\sqrt{6}} \right), \frac{e^\chi}{2} + e^{-\chi} \left(\frac{\cos(\theta)}{2} + \frac{\sin(\theta)}{\sqrt{6}} \right) \right)$$

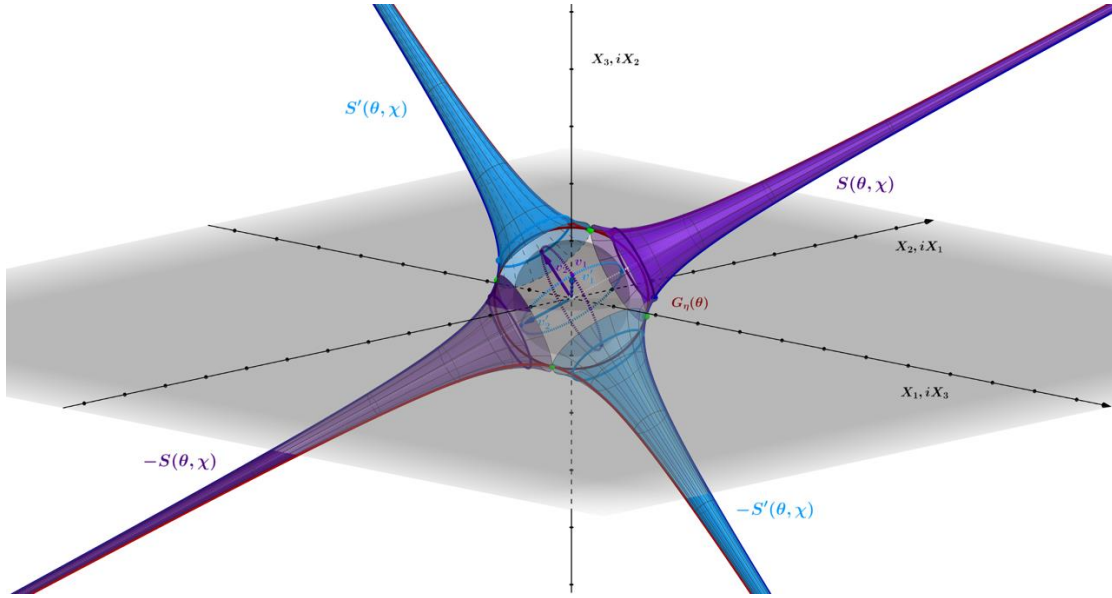


Figure 8. – The set of four Torricelli trumpets $\{\pm S(\theta, \chi), \pm S'(\theta, \chi)\}$ oriented around the unit sphere.

In **Figure 8.**, we present the set of four Torricelli trumpets $\{\pm S(\theta, \chi), \pm S'(\theta, \chi)\}$ oriented around the unit sphere. Geometrically, the Light Glowney-Dirac Delta formalism defines a rotating circular boundary embedded within the curvature field of the Polyplex manifold. Its fixed centroid displacement from the origin represents a symmetry-broken reference frame, anchoring observer dynamics at a constant offset from the modular center of the MegaCube.

This construction gives rise to the so-called Torricelli trumpet projection, wherein the circular frame rotates continuously while preserving radial eccentricity. The varying radius imposes a phase-asymmetric expansion along the light-shell axis, generating a trumpet-like surface. This surface captures the graded boundary between symmetry and asymmetry, between spherical containment and hyperbolic divergence.

Physically, the Light Glowney-Dirac Delta defines a localized observer frame that measures field content across the trumpet's curvature. Its circular motion introduces an azimuthal degree of freedom into the Polyplex geometry, allowing

the Third Observer to define angular projection mappings for Lie algebra observables. As a whole, the geometry is not isotropic, revealing preferred curvature orientations embedded in the metric field.

Topologically, the system embeds four such trumpets arranged tetrahedrally around the unit sphere, forming a cardinal geometry of observational symmetry. Each trumpet represents a projection of the Dirac delta onto a curved boundary, modulated by rotational phase θ , and together these projections span the curvature domain of the Polyplex manifold's light structure.

The Light Glowney-Dirac Delta formalism introduces a structured asymmetry into the observer field, creating geometric locality and directional curvature coupling. The Third Observer thus encodes phase-locked reference structure, bridging the continuous Hilbert manifold with the modular quantization of the MegaCube and grounding angular motion into a radial-hyperbolic framework of directional measurement.

XII. The Singularity's Ellipsoidal Transform

The Singularity's Ellipsoidal Transform defines a critical topological operation within Polyplex Lie Geometry, encoding how bounded elliptical structures are embedded and rotated through the trumpet curvature fields surrounding the central singularity. These transformations articulate the continuous deformation of geometric surfaces around the light cone and introduce an eccentric angular structure into the surrounding space.

A slice of the trumpet surface is defined by the two hyperbolic orthogonal equations:

$$\begin{aligned}\eta_{+,+} &= (\cosh(\ln(t)), e^{\ln(t)}, \sinh(\ln(t))), \quad t \in \mathbb{R}, \\ \eta_{-,-} &= (\sinh(\ln(t)), e^{\ln(t)}, \cosh(\ln(t))), \quad t \in \mathbb{R},\end{aligned}$$

These describe orthogonal sections through a trumpet-like geometry, where hyperbolic sine and cosine functions generate asymmetric expansion along their respective coordinates. This expansion defines the asymptotic curvature envelope of the singularity.

The following elliptic function is always circumscribed by the boundary of the asymptotic light region:

$$r(\theta, \chi) = (0,0,0) + \left(\cos(t)\cosh(\chi) + i\sin(t)\sinh(\chi), (\cos(t) + i\sin(t))e^\chi, \cos(t)\sinh(\chi) + i\sin(t)\cosh(\chi) \right)^T,$$

which, for any given variable values, will have four intersections with a trumpet and its antipodal analog trumpet. These intersections define symmetry-breaking junctions where curvature transitions from hyperbolic divergence to elliptical confinement.

Transforming this into a real-valued parametric form, we write:

$$\rightarrow r_{\mathbb{R}}(\theta, \chi) = \begin{pmatrix} X(\theta, \chi) \\ Y(\theta, \chi) \\ Z(\theta, \chi) \end{pmatrix} = \begin{pmatrix} \cos(\theta)\cosh(\chi) + \sin(\theta)\sinh(\chi) \\ (\cos(\theta) + \sin(\theta))e^\chi \\ \cos(\theta)\sinh(\chi) + \sin(\theta)\cosh(\chi) \end{pmatrix}, \quad \theta \in [0, 2\pi], \{t, \chi\} \in \mathbb{R}$$

This curve describes a 2D ellipse, which, when rotated along a spatial axis, forms the surface of an ellipsoid embedded within the trumpet domain.

The ellipsoid rotation axis is aligned along the direction of the semi-major axis $\hat{u}$:

$$\hat{u} = \left(\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right)$$

This vector represents a diagonal symmetry in the complexified Polyplex space and projects the ellipsoid along an axis that intersects the MegaCube at discrete rational coordinates. The ellipsoid will have a semi-major axis equal to $\sqrt{3}e^\chi$, and two equal semi-minor axes equal to $e^{-\chi}$. We use Rodrigues' rotation formula to the base vector $r_{\mathbb{R}}(\theta, \chi)$, producing the fully rotated surface.

$$R(\theta, \phi, \chi) = r_{\mathbb{R}}(\theta, \chi) \cos(\phi) + (\hat{u} \times r_{\mathbb{R}}(\theta, \chi)) \sin(\phi) + \hat{u}(\hat{u} \cdot r_{\mathbb{R}}(\theta, \chi))(1 - \cos(\phi)),$$

We arrive at the following component-wise parametric forms:

$$\begin{aligned}X'(\theta, \phi, \chi) &= \frac{e^{-\chi}(3 \cos(\phi) - \sqrt{6} \sin(\phi))(\cos(\theta) - \sin(\theta))}{6} + \frac{e^\chi(\cos(\theta) + \sin(\theta))}{2}, \\ Y'(\theta, \phi, \chi) &= \frac{e^{-\chi} \sin(\phi) (\cos(\theta) - \sin(\theta))}{\sqrt{6}} + e^\chi(\cos(\theta) + \sin(\theta)),\end{aligned}$$

$$Z'(\theta, \phi, \chi) = -\frac{e^{-\chi}(3 \cos(\phi) + \sqrt{6} \sin(\phi))(\cos(\theta) - \sin(\theta))}{6} + \frac{e^{\chi}(\cos(\theta) + \sin(\theta))}{2}.$$

This form captures the full ellipsoidal transform of the singularity's surface, with eccentric curvature expanding and compressing through the trumpet shell, producing a directional ellipsoid. This surface embeds the dual geometric behavior of curvature symmetry and exponential scaling.

The embedded ellipsoid acts as an angular mediator between symmetric spherical behavior and hyperbolic extension, introducing a bounded rotational domain inside an otherwise asymptotically diverging geometry. Its intersections with the trumpet and antipodal trumpet form a quadripartite spatial junction, encoding symmetry operations through geodesic curvature.

From a physical standpoint, this ellipsoidal transform can be interpreted as a compressed quantum harmonic domain bounded by hyperbolic walls, analogous to the energy quantization seen in elliptical potential wells. It defines phase-bounded metric regions that preserve angular momentum in the presence of exponential curvature.

As the ellipsoidal transform aligns to the *MegaCube's* axis of rational intersection, it additionally encodes a lattice-coherent reference frame, suitable for modeling spinor fields, modulated curvatures, and discrete phase rotation in a unified algebraic setting. In this way, the singularity's ellipsoidal transform serves as a bridge between curvature-bound differential geometry and lattice-encoded modular symmetry, enriching the Polyplex Lie Geometry landscape with phase-contoured ellipsoidal microdomains.

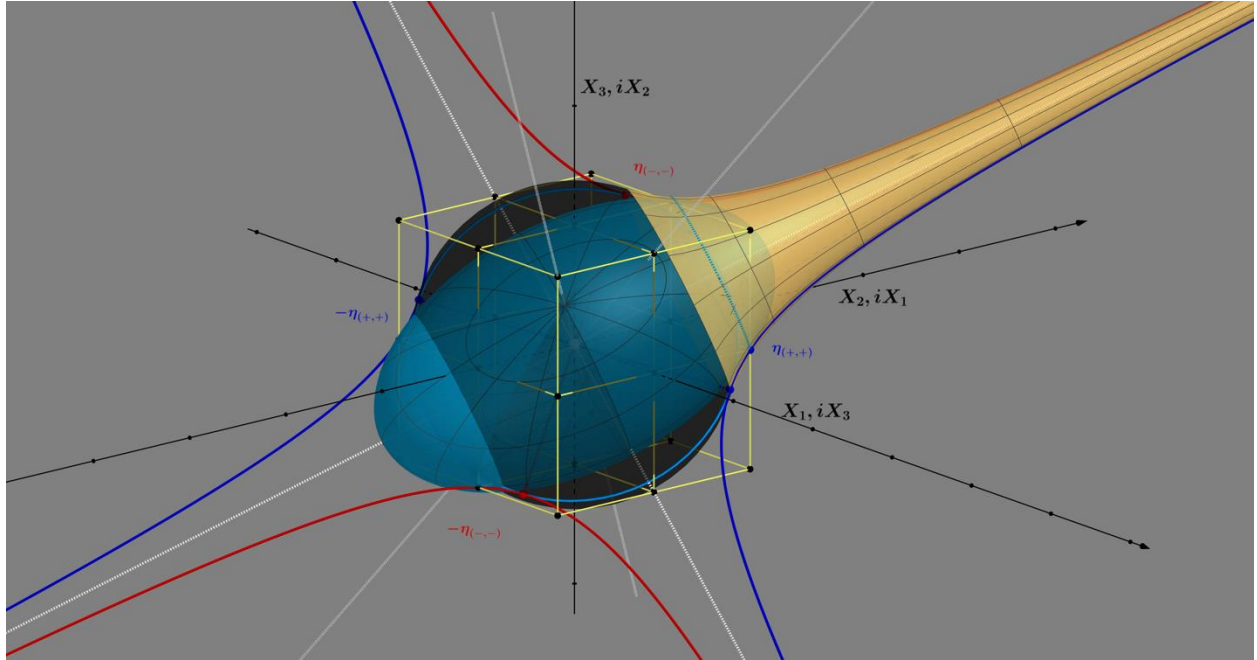


Figure 9. – The ellipsoidal transform (light blue) of singular sphere (black), and its intersection with the Toricelli trumpet (gold) along the Q^+ axis (dotted light blue circle).

XIII. Conclusion

Polyplex Lie Geometry presents a unified geometric and algebraic framework that reconciles continuous field-theoretic symmetries with discrete lattice-structured modularity. It begins with a foundational singularity, expands outward into layered curvature manifolds, and inward into algebraic minimality, all embedded within a complex Hilbert space structured by geometric and Lie algebras. The fusion of spherical, hyperbolic, elliptic, and lattice geometries into one continuous-differentiable and combinatorially modular system enables a new language of geometry that encodes quantum observables, gauge fields, topological states, and symmetry-breaking events with structural coherence.

At its core, Polyplex Lie Geometry captures not just the metric diversity of space but the algebraic and transformational grammar of field evolution, embedding Lie algebra representations, subtangent projections, and infinite spheres of tangent spaces within a higher-order operator algebra. The interplay between the Angle Space, the *Infinite Sphere of*

Tangent Spaces (ISTS), and the *MegaCube* lattice defines a multi-layered topology that is simultaneously quantum-coherent, topologically rich, and geometrically expressive.

Structures such as the elliptic group $\mathcal{E}_\eta(\theta, \chi)$, the Light Glowney-Dirac Delta, and the Singularity's Ellipsoidal Transform illustrate how curvature, phase, and modular quantization can interact dynamically to generate rich submanifolds, directional observables, and localized symmetry frames. Each construct encodes a different facet of the Polyplex principle: a layered, algebraically-driven geometry that supports dualities between bounded and unbounded motion, isotropic and anisotropic curvature, and discrete and continuous transformational behavior.

Through this framework, geometry is no longer a passive container of structure but an active, algebraically-defined manifold of interactions. It admits dynamic modulation through subtangent flows, topological quantization via lattice intersection, and physical interpretation through harmonic embedding and thermodynamic partition. The Polyplex Lie Geometry thus proposes a rich mathematical environment for encoding not only geometry and symmetry, but the deeper architecture of physical law and the potential structure of consciousness itself.

In unifying angle-based manifolds, modular algebras, operator geometries, and discrete topologies, Polyplex Lie Geometry lays the groundwork for future investigations into gauge-gravitational unification, algebraic cosmology, quantum field curvature, and observer-relative differential structure. It is a geometry of projection and synthesis, of singularity and emergence, where the continuous and the discrete cohere in a higher-order algebraic space structured by curvature, quantization, and rotational symmetry.